Do Now · retrieval

Testing and program quality

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why should Largest() require a nonempty input array?

Q2

What is a regression?

Q3

What does a class method do?

Q4

A date validator passes all unit tests but booking fails when writing to the database. What test is needed?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Test individual units — Unit tests isolate one function or class method from the rest of the system.

Core idea 2

Test connections and behaviour — Integration tests interactions between modules; system tests the complete application.

Core idea 3

Confirm intended outcome — Acceptance testing checks whether the delivered application meets stakeholder requirements. Automated regression tests rerun after changes.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

unit testing — Test one module/function in isolation.

integration testing — Check interaction of components.

system testing — Test the entire application as a system.

acceptance testing — Check fitness for user/business requirements.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

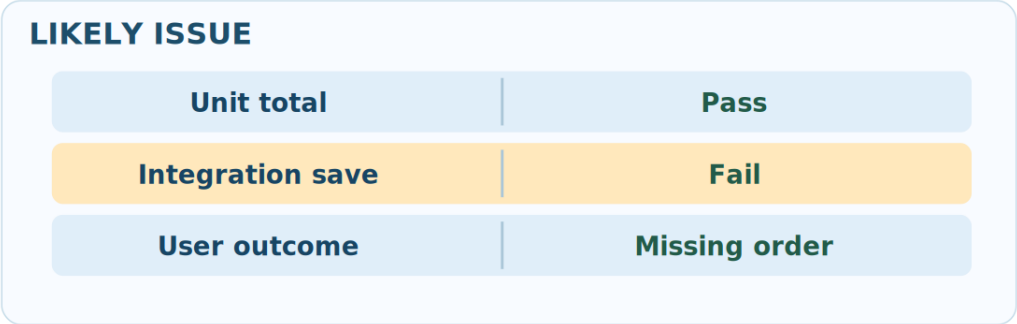
A checkout computes the right total but fails to save orders. Which test is most relevant?

Follow the worked method

1. Predict first — TotalCalculator already gives correct values; the failure appears when OrderService interacts with the database.

2. Model and explain — Prioritise an integration test of saving, then a system-level end-to-end checkout test.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
The isolated search passes, so check the connection that maps returned IDs to displayed titles.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Match each: individual ValidDate function, database interaction, full website journey, client sign-off to the appropriate test level.

Question 2

A login test passes until a new password rule is introduced, after which registration fails. Explain a sensible debugging and testing workflow.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Why is 100% execution of code lines not equivalent to proving an application correct? Suggest three further forms of evidence.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Passing unit tests alone proves that the complete application works for its users.

Use key vocabulary from memory

1. Explain unit testing in your own words.

2. Explain integration testing in your own words.

3. Explain system testing in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.